

Criteria2Query 3.0: Leveraging Generative Large Language Models for Clinical Trial Eligibility Query Generation

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ABSTRACT

Objective: Automated identification of the eligibility of patients is a bottleneck of prescreening in clinical research. We propose Criteria2Query (C2Q) 3.0, a system that leverages GPT-4 for semi-automatic parsing of clinical trial eligibility criteria to identify suitable patients within clinical databases.

Materials and Methods: C2Q 3.0 integrated three GPT-4 prompts for concept extraction, SQL query generation, and reasoning. Each prompt was designed and evaluated separately. The concept extraction prompt was benchmarked against manual annotations from 20 clinical trials by two evaluators. For SQL generation accuracy, two evaluators identified errors in GPT-generated SQL queries from 5 clinical trials. The reasoning prompt was assessed by three evaluators in four aspects: readability, correctness, coherence, and usefulness, using corrected SQL queries and a feedback questionnaire.

Results: Out of 518 concepts from 20 clinical trials, GPT-4 achieved an F1-score of 0.891 in concept extraction. In SQL generation, 29 errors spanning seven categories were detected, with logic errors being most common ($n = 10$; 34.48%). Reasoning evaluations yielded a high coherence rating (mean score of 4.70), but relatively lower readability with a mean of 3.95. Mean scores of correctness and usefulness were identified as 3.97 and 4.37, respectively.

Discussion and Conclusion: Incorporating GPT-4, C2Q 3.0 significantly improves the processing of eligibility criteria and patient identification for clinical trials, while upholding data privacy and reducing the need for frequent updates. Continued research is necessary to solidify the reliability of large language models within the system.

INTRODUCTION

The recent advances in generative large language models (LLMs) such as ChatGPT and its subsequent model GPT-4 have shown remarkable performances in language understanding and generation. [1, 2] A massive amount of textual data and model parameters enable the creation of comprehensive language models, resulting in human-like text generation. [3-6] Moreover, these LLMs have demonstrated their abilities in not only language generation, but also inference, knowledge mining, and logical reasoning. [7-10] However, there has been limited evidence of the usage and effects of LLM in parsing and processing the eligibility criteria of clinical trials. Hamer et al. evaluated the screenability of the eligibility criteria using LLM for 4,135 criteria across 146 clinical trials and found that 72% of the criteria were correctly identified by using LLM. [11] Wong et al. evaluated GPT's ability to structure the eligibility criteria of 53 oncology trials and reported that GPT-4 achieved F1-score from 64.8 to 65.4 on histology identification and 66.7 to 72.5 on biomarker identification. [12] Meanwhile, Peikos et al. leveraged ChatGPT to extract patient-related information from clinical notes and to match it with clinical trials. [13] Jin et al. developed TrialGPT which employed GPT-3.5 to parse eligibility criteria and derive relevance scores on the criterion-level from given patient information. [14] TrialGPT was evaluated on three publicly available cohorts of 184 patients and 18,238 annotated clinical trials, and outperformed the BERT-based models for ranking and excluding trials. These studies directly match patient records with clinical trials in GPT prompt. Even with impressive performance, the feasibility of the systems is far from practical use-cases as GPT is only available via online OpenAI API and patient private data cannot be transmitted to non-authorized third party in most cases. [15, 16] Furthermore, as patient records in clinical databases accumulated over time, patient-trial matching systems need to be re-executed periodically. However, the proposed methods require substantial computing

resources and the expense of using OpenAI API, and even might not be able to reproduce the result as GPT performance changes. [17] Therefore, a different approach is required to ensure efficiency and reproducibility while preserving patient records safely.

Criteria2Query (C2Q) 1.0 is a piece of open-source software for automating information extraction and query formulation directly from free-text clinical trial eligibility criteria. [18] It combined machine-learning and rule-based methods to systematically parse criteria and convert them into SQL queries conforming to the Observational Medical Outcomes Partnership (OMOP)-Common Data Model (CDM). [19] C2Q 2.0 extended C2Q 1.0 by involving humans in the loop for more accurate concept recognition, entity and value normalization, and criteria simplification. [20] Despite the efforts previously made, both systems still have unsatisfying parsing results due to complex logical relationships among concepts. [21-23] For example, C2Q 2.0 can recognize only three concepts ('renal disorders', 'cardiovascular', and 'immune disease') from the sentence "hepatic or renal disorders or cardiovascular, endocrine, metabolic, immune diseases". In addition, the current method of C2Q 2.0 struggles to parse criteria comprising multiple measurements and respective upper and lower thresholds within a single sentence. For example, the thresholds for the measurements from "systolic blood pressure ≥ 140 and < 180 mmHg and diastolic blood pressure ≥ 90 and < 110 mmHg" are hard to extract. These textual representations are complicated to process but are common in eligibility criteria. Furthermore, the current OMOP's query formulation approach is rule-based. While accurate, it requires massive entities and subqueries, resulting in inefficient query execution time. [24]

We hypothesize that comprehensive language understanding and generation abilities of LLM can overcome these challenges. While previous studies have indicated the potential of GPT-based frameworks in matching eligibility criteria with patient records, the implementation of such approaches in real-world settings remains challenging due to data privacy concerns and the required periodic updates in patient records. In this paper, we present C2Q 3.0 incorporating OpenAI GPT-4, with a primary focus on enhancing the information extraction and query formulation processes. We aimed to facilitate efficient patient pre-screening processes while maintaining data privacy and minimizing resource consumption. In addition, from our experience learned from real-world user feedback, we intended to enhance system interpretability by providing users with a narrative description of the system processing procedure. Herein, our contribution includes 1) comparing GPT's parsing performance on complex eligibility criteria with the existing method, 2) identifying potential uses and errors of GPT-generated queries to enhance query performance, and 3) implementing a user interface to interpret generated SQL query and standardized OMOP concept to enhance human-computer interaction.

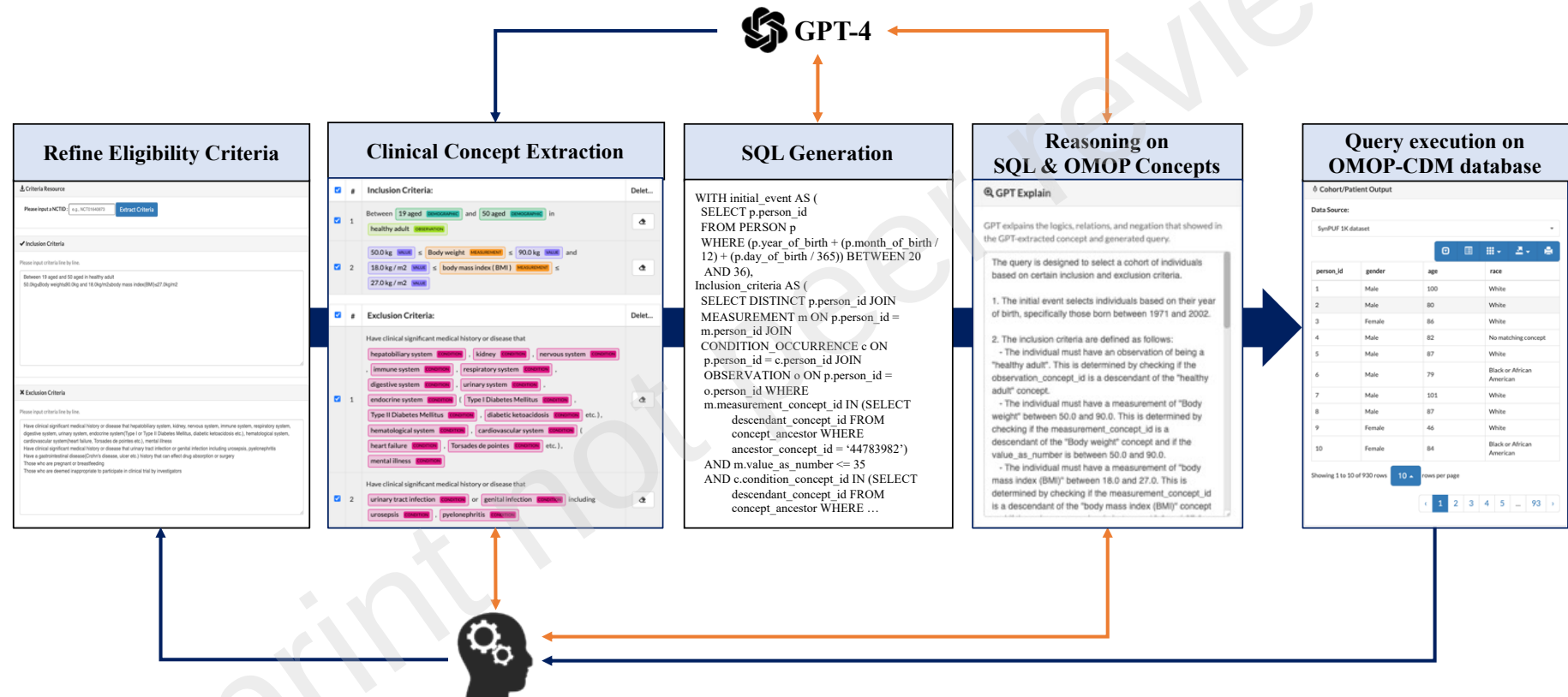
METHODS

System Overview

The overall workflow of C2Q 3.0 is presented in Figure 1. We have implemented three GPT-prompts in the system: 1) concept extraction, 2) SQL generation, and 3) reasoning. OpenAI GPT-4 API was used in the system and the temperature was set to 0.0 for reproducibility. [2] We adopted one-shot prompting in all prompts based on its demonstrated accuracy and consistency across a variety of use cases. [25] Once the system receives the eligibility criteria, it uses GPT to recognize

the clinical entities and leverages Usagi to map the entities to standardized concepts in OMOP-CDM. [19] Usagi is an automated concept-mapping software provided by Observational Health Data Sciences and Informatics (OHDSI). [26] The user-editable interface was inherited from the previous versions of C2Q, allowing criteria selection, parsing error correction, and concept mapping. After concept mapping, the system formulates SQL queries and reasoning explanations behind the query and mapping. The outputs of the parsing, mapping, and reasoning processes are displayed with a user-editable interface. The interface allows editing the GPT-parsed clinical concepts, Usagi-mapped standardized concepts, and GPT-generated reasoning on formulated SQL query. The editing of the reasoning explanation can be sent to GPT and used for re-formulating SQL query. After user modifications, the system re-generates the consequent SQL query and reasoning, and the outputs (both query and reasoning) are refreshed subsequently. Once the SQL query is confirmed by the user, the system executes the query to the OMOP-CDM formatted database and returns the cohort information.

Figure 1. Overall workflow of Criteria2Query 3.0 powered by GPT-4. The orange colored-arrows represent the interaction between human or GPT. If the users modify GPT-generated results, GPT will recognize the changes and re-execute each corresponding module.



Prompt design

Concept extraction

Figure 2A shows the overall prompt design for concept extraction. We utilized the prompting practices from previously proposed strategies. [25] In the concept extraction prompt, we described information extraction and normalization functionalities in the task description section. The functionalities include named entity recognition, negation detection, abbreviation normalization, and attribute normalization. [18, 20] All entities were extracted and assigned to the categories in OMOP-CDM: Demographic, Condition, Drug, Device, Procedure, Measurement, and Observation. Additionally, we added Visit, Temporal, and Value categories that can be considered as the attributes of OMOP categories. To provide more flexibility in parsing complex criteria, we allowed GPT to create new categories for the recognized medical entities to better structure information. For example, the entities such as ‘cycle of chemotherapy’ and ‘left/right site of kidney’ cannot perfectly fit into any predefined categories above, GPT assigned ‘drug cycle’ and ‘anatomic site’, respectively. We also provided the definition of each category in OMOP-CDM tables that is described in OMOP-CDM specifications [27] and the input and output data specifications in the prompt.

SQL generation

We implemented a text-to-SQL approach to generate SQL queries to retrieve the patient cohort conforming to the eligibility criteria. Figure 2B presents the design of the SQL generation prompt. We adapted the existing prompt designs for text-to-SQL generation to our OMOP-specific SQL query generation. [28, 29] For example, we provided and taught GPT to use standardized OMOP concept IDs while generating the query. The task description also includes

considering the relations and logic lying in the eligibility criteria. To diminish syntax errors, we defined the output format to conform to the SQL dialect. As a one-shot prompting, an example query is provided. Information about what categories of entities can be found in each of the OMOP-CDM tables is also provided in the prompt to enhance the accuracy in identifying the appropriate tables for search records of clinical concepts. For example, Demographic information can be extracted from the PERSON table and Condition can be extracted from the CONDITION_OCCURRENCE table.

Reasoning

This prompt aims to improve the system interpretability by providing an explanation to the users of how the system generates queries from the given eligibility criteria. Specifically, the prompt focuses on explaining the relations between concepts, logic between the eligibility criteria, and values or units of lab measurements, which cannot be identifiable in the current user interface. Since we adopted the GPT-based SQL generation process, the generated SQL query could have errors and biases, requiring a human-in-the-loop process to minimize the selection bias while pre-screening patients. Particularly for individuals lacking knowledge in data structures, SQL, or the OMOP-CDM, the human-like language generation of GPT can facilitate understanding of the system processing on parsing and queries. [30] Similar to the other prompts, we adapted the preceding prompting strategy, incorporating OMOP-specific knowledge. [31] The task was set to narratively describe the logic, relation, negation, and other attributes of the observed concepts in the query (Figure 2C). In this prompt, GPT-generated SQL query and extracted entities with standardized OMOP concept-IDs are provided as input

data. The output was set as a narrative explanation with numbered paragraphs to enhance readability.

Figure 2. Example architectures of three prompts in Criteria2Query 3.0

A) Concept Extraction

Task description	Annotate clinical concepts from the given text using the following rules: 1) Annotate concepts with ...
Output format specification	Return your answer with extracted entities and under [Annotation] section.
Output example	Following is allowed examples: 1) <Measurement>eGFR</Measurement> <Value>triple negative</Value>
OMOP schema information	Condition is events of a Person suggesting the presence of a disease or medical condition stated ...
=====	
Output	[initial_event] [inclusion_criteria] Patients diagnosed with <Condition>hypertension</Condition>

B) SQL Generation

Task description	Generate PostgreSQL query from the given text while following the rules: 1) Target database schema is OMOP ...
Input format specification	The given text consists of three sections: [initial_event], [inclusion_criteria], and [exclusion criteria] ...
Output format specification	Use Common Table Expression and add semi-colon after each sql statement.
Output example	WITH initial_event AS (SELECT ...your query...), inclusion_criteria AS (SELECT ...your query...), ...
=====	
Output	WITH initial_event AS (SELECT person_id from PERSON where gender_concept_id = 8532), inclusion_criteria

C) Reasoning

Task description	Explain logics, temporal, relation, and attributes between annotated concepts from a given query ...
Input format specification	The query confirms to OMOP-CDM v5.3.2 and the concept mapping list consists of [data-entity], [concept-id] ...
Output format specification	Return your answer as the narrative explanation with numbers. Explain the concepts with [concept-name] ...
Output example	1. The initial_event is ...your answer... 2. The inclusion_criteria is defined as ...your answer...
=====	
Output	The given SQL query aims to identify a cohort of patients based on inclusion and exclusion criteria. The initial ...

Evaluation

The tasks were evaluated independently with clinical trials collected from ClinicalTrials.gov (www.ClinicalTrials.gov). We purposely selected twenty clinical trials covering a variety of conditions, interventions, and trial phases to perform a comprehensive and rational evaluation. Trials with improper format style, vague clinical abbreviation, or which do not include both inclusion and exclusion criteria were excluded. For example, trials that do not distinguish inclusion or exclusion criteria are excluded, and trials including abbreviations without definitions such as FC (potentially meaning fevers and/or chills, functional class, etc.) or HSS (potentially meaning hyperosmolar hyperglycemic state or hepatic stimulator substance) are excluded. To measure the evaluation adequacy of the trials, we calculated the complexity of the eligibility criteria which measures the number of complex patterns of clinical concept in a single phrase, and trimmed the trials with complexity scores less than 0.1 (extremely simple) or above 0.9 (extremely complex). [32] Overall, twenty clinical trials of 18 diseases were selected and the average complexity of the selected trials was identified as 0.46 (Supplementary Table S1). Five clinical trials whose free-text eligibility criteria were successfully translated to OMOP-CDM (NCT05880693, NCT05719155, NCT05521633, NCT05535595, and NCT05479214) were used for the evaluation of the SQL query generation and GPT-reasoning. Only five trials were evaluated here due to the extensive effort required for these tasks.

Overall, seven evaluators participated in the evaluation. Concept extraction and SQL generation tasks were each evaluated by two biomedical informaticians, and the reasoning task was evaluated by three clinicians. For concept extraction, reviewers were instructed to annotate

clinical concepts with OMOP categories, and their consensus results were considered the gold standard and were compared with the system-extracted concepts. Cohen's kappa coefficient was calculated to assess the inter-rater agreement. In addition, we compared the concept extraction performance of the current system (GPT-4) with GPT-3.5-turbo and with C2Q 2.0. Precision, recall, and F1-score are used as evaluation metrics. For SQL generation, evaluators first manually reviewed the GPT-generated queries to identify semantic errors (e.g., criteria misinterpretation, function misuse, or concept omissions) and executed them on the OMOP-CDM database to confirm the validity of the queries. To evaluate GPT-generated reasoning, we developed a 17-item questionnaire consisting of four categories: readability; correctness; coherence; and usefulness. The evaluators were instructed to compare the target free-text eligibility criteria and GPT-generated reasoning and to score the questions. The items were rated using a 5-point Likert scale, ranging from "Strongly disagree" as (1) to "Strongly agree" as (5).

RESULTS

Concept extraction

Overall, 518 entities were identified from twenty clinical trials by two evaluators (Supplementary Figure S1). In our study, the inter-rater agreement measured by the kappa coefficient achieved a substantial agreement of 0.75. [33] The most frequent entity category was Condition, with a total of 206 (39.77%) occurrences. Value and Observation were the following most frequent categories, accounting for 78 (15.06%) and 59 (11.39%) occurrences, respectively. On the other hand, Device (1; 0.19%) and Visit (2; 0.39%) were identified as the least frequent categories in all criteria.

Table 1 presents a comparative analysis of the performance among the three systems: C2Q 3.0 with GPT-4, C2Q 3.0 with GPT-3.5-turbo, and C2Q 2.0. C2Q 3.0 with GPT-4 resulted in the highest scores across all evaluation metrics, with a precision of 0.863, recall of 0.922, and F1-score of 0.891. Meanwhile, C2Q 3.0 with GPT-3.5-turbo achieved the lowest recall (0.534) and F1-score (0.612).

Table 1. Performance comparison on concept extraction among different versions of the Criteria2Query system.

System	Precision	Recall	F1-score
Criteria2Query 3.0 with GPT-4	0.863	0.922	0.891
Criteria2Query 3.0 with GPT-3.5	0.717	0.534	0.612
Criteria2Query 2.0	0.667	0.753	0.707

SQL generation

Overall, 29 errors with seven different error types were identified. The specific error types and their definitions are presented in Table 2. We classified three types of semantic error as logic error, concept omission, and incorrect concept mapping. Logic error was defined as the error related to the misinterpretation of the relation, temporal, quantity, and negation expression in the criteria. For example, misinterpretation of age criteria or switching the values between two different lab measurements were classified as logic errors. Concept-related errors were defined as the concept missing which identified concepts in the concept extraction process but missed in SQL formulation, and the incorrect concept mapping. Incorrect concept mapping resulted from Usagi, the adopted framework for concept mapping. The other errors (i.e., functional

misuse, integrity constraint violation, schema reference error, and syntax error) were structural errors, related to the foundational structure of the SQL query. The distribution of the errors across the five clinical trials is presented in Figure 3. Logic error was identified as the most prevalent error with 10 (34.48%) occurrences, followed by concept missing and incorrect concept mapping errors, each with 6 (20.69%) occurrences respectively. The most common logic error was identified as duplicated patient records in the cohort ($n = 5$), resulting from missing the DISTINCT statement in the query. In contrast, integrity constraint violation and schema reference error were identified as the least frequent errors, both with 1 (3.45%) occurrence. Meanwhile, the clinical trial (NCT05521633) whose eligibility criteria achieved a relatively high complexity score (0.64), had the most errors ($n = 10$). Conversely, the trial (NCT05719155) whose criteria were with a low complexity score (0.29), had the fewest error ($n = 2$).

Figure 3. Manually identified SQL errors from GPT-generated queries on five clinical trials

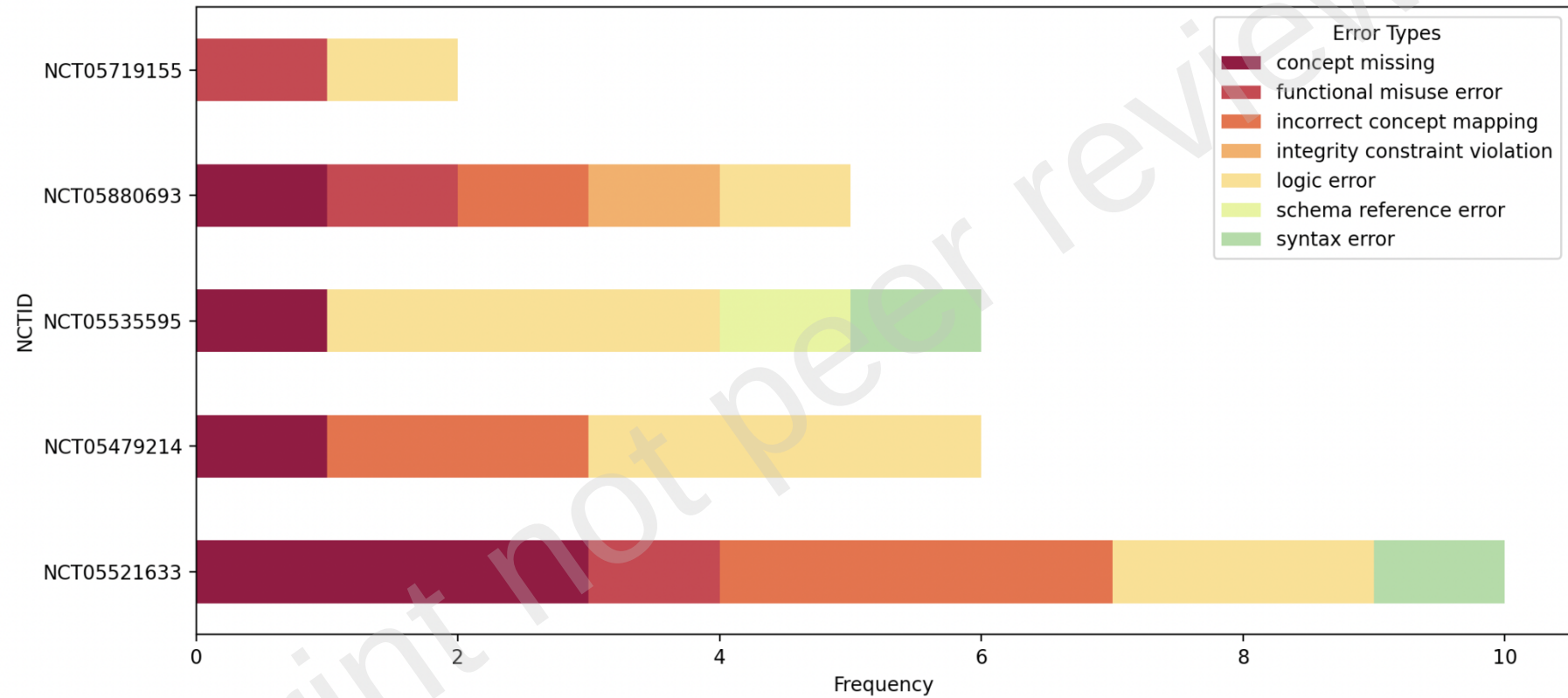


Table 2. Identified error types and definitions from GPT-generated SQL query.

Error type	Definition
Concept missing	Concept missing error means that a clinical concept is detected at the concept extraction stage but not used in query formulation.
Functional misuse error	Functional misuse error means that SQL functions are incorrectly used in the query and eventually cause the error.
Incorrect concept mapping	Incorrect concept mapping means that extracted clinical concepts are mapped to the wrong standardized OMOP vocabulary.
Integrity constraint violation	Integrity constraint violation means that the query violates the OMOP's data consistency such as inserting string value to the columns expecting integer concept IDs.
Logic error	Logic error means that GPT incorrectly defines the logic, relation, or temporality of the clinical concepts in the query.
Schema reference error	Schema reference error means that GPT incorrectly referenced OMOP-CDM tables or columns.
Syntax error	Syntax error is an error that has incorrect SQL syntax such as misspelled words or incorrect use of punctuation in the query.

Abbreviations: OMOP, the Observational Medical Outcomes Partnership; CDM, common data model

Reasoning

Table 3 and Supplementary Figure S2 present the evaluation of the reasoning task based on four main categories: readability, correctness, coherence, and usefulness. The mean score of readability questions was 3.95 (standard deviation [SD] = 0.64). The clarity of the initial events, inclusion, and exclusion criteria of the trial was rated ranging from 3.22 to 3.53, which suggests GPT's understanding of initial event, inclusion, and exclusion criteria can be relatively accurate. The clarity of layout, writing format, concepts, and terminologies were rated ranging from 4.33 to 4.80. The mean score of correctness was 3.97 (SD = 0.58). The appropriate use of clinical terminologies and concepts was scored as 3.60 (SD = 0.80), and the completeness of the concepts was scored as 3.33 (SD = 0.90), meaning some of the clinical concepts in the protocol are not mentioned in GPT-generated reasoning which is caused by the concept missing error in SQL query. Meanwhile, the logic, relation, and attribute between concepts were rated ranging from 4.40 to 4.60. The mean score of coherence was 4.70 (SD = 0.14). Coherence was scored highly, particularly on the clear explanation of the relations between the different criteria of the trial. Lastly, the mean score of usefulness was 4.37 (SD = 0.52), indicating that the reasoning prompt aids in understanding the system and the clinical trial, as well as being a useful resource for modifying or reproducing the query. The only aspect scoring lower in usefulness was the presentation of the OMOP concepts which was rated as 3.67 (SD = 0.95).

Table 3. Questionnaire and mean scores on GPT-generated reasoning explanation

Category	Question	Mean score (SD)
Readability	The answer clearly explains the initial event, if exists, of the clinical trial	3.22 (1.02)
	The answer clearly explains the inclusion criteria of the clinical trial	3.53 (0.70)
	The answer clearly explains the exclusion criteria of the clinical trial	3.40 (0.69)
	The clinical concepts and terminologies are easy to understand.	4.33 (0.50)
	The layout of the answer is easy to follow, such as numbered points and bullet points.	4.40 (0.53)
	Overall writing-format is easy to understand.	4.80 (0.35)
Correctness	The answer accurately uses the appropriate clinical terminologies and concepts.	3.60 (0.80)
	All clinical concepts mentioned in the trial protocol are mentioned in the explanation.	3.33 (0.90)
	The logics between concepts are clearly explained.	4.60 (0.35)
	The relations between concepts are clearly explained	4.53 (0.12)
	The attributes (e.g., values of measurement) between concepts are clearly explained	4.40 (0.40)
Coherence	The relations between initial, inclusion, and exclusion criteria are correctly explained.	4.80 (0.20)
	The answer consistently maintains the same level of detail when explaining the different parts of the query.	4.60 (0.53)
Usefulness	The answer is helpful to understand how the system understand the clinical trial and generated the query.	4.73 (0.31)

	The answer is useful if you want to modify or reproduce the query.	4.27 (0.70)
	Presenting the OMOP concepts used in the query is helpful to understand which patients are queried.	3.67 (0.95)
	Explaining the logic, relation, and attributes between concepts are helpful to understand which patients are queried.	4.80 (0.35)

Abbreviation: OMOP, the Observational Medical Outcome Partnership; SD: standard deviation

DISCUSSION

In this study, we developed C2Q 3.0 leveraging OpenAI GPT-4 to enhance accurate criteria parsing and query formulation from the free-text eligibility criteria. We also introduced the GPT-generated reasoning prompt to interpret the SQL query and standardized concepts to the users. C2Q 3.0 aimed to automate the overall process from parsing the free-text eligibility criteria to generating SQL query for a standardized clinical database. Our proposed system design can accurately match patients with clinical trials while protecting data privacy since GPT was leveraged only for parsing the eligibility criteria and generating SQL query, not directly matching trials with patient data. The generated queries can be executed on the user's clinical database; therefore, no patient data is transferred to OpenAI. Our system incorporated three different prompts: concept extraction, SQL generation, and reasoning. The results demonstrated that GPT-4 outperformed the C2Q 2.0 and GPT-3.5-turbo in criteria parsing. The most prevalent errors of GPT-generated queries were logic and concept-related errors, and complex criteria tend to generate more errors. While GPT's interpretation of the generated SQL queries is far from perfection, it still showed promising results and was found to be potentially useful for users to modify or reproduce the query.

In concept extraction, GPT-4 showed the highest performance in all evaluation metrics. Upon conducting an error analysis, we observed that most of the false positive clinical concepts of GPT-4 (e.g., "English speaking", "patient's willingness", or "patient's agreement") were deemed to be clinical entities that are not queryable in OMOP databases, and these false positive cases might lower the precision. These concepts are easily comprehensible for humans, but require extensive rules to process in the machine or are not representable in the OMOP-

CDM. [34, 35] Further studies should be conducted to explicitly distinguish between queryable and non-queryable clinical concepts in the criteria. Meanwhile, GPT-3.5-turbo showed lower performance and even failed to conform to the output format instruction in the prompt, consistent with the results from other studies (Supplementary Figure S1). [15, 25, 36-38] The studies also suggested that traditional language models fine-tuned on a specific domain can achieve higher performance than GPT. [25, 39, 40] However, GPT-4 outperformed the C2Q 2.0 which leverages traditional language models trained on clinical trial eligibility criteria corpus. [18, 20] The performance of LLMs in biomedical research is still controversial and requires further evaluation in several aspects. [41] We identified that C2Q 3.0 with GPT-4 can accurately recognize concepts in complex criteria, compared to GPT-3.5-turbo or C2Q 2.0 (Supplementary Figure S3). For the trial NCT05719155, while C2Q 2.0 could not distinguish between serially mentioned measurements and their corresponding values, GPT-4 correctly identified each Measurement entity. GPT-3.5-turbo could recognize the exact spans of the entities, but it failed to annotate the correct OMOP category. In NCT05532800, C2Q 2.0 was unable to identify CABG, while GPT-4 was able to identify all entities and was able to perform abbreviation normalization. However, the system using GPT-3.5-turbo was not adhering to the specified output format and failed to recognize certain entities with a correct OMOP category.

The errors identified from GPT-generated SQL queries were mostly related to logic and concepts, critical for accurate patient retrieval. Since GPT is maintained by OpenAI, we were not able to identify the cause of the errors in this study. However, one possible reason for the errors is the performance variation given the length of the prompt context. [42, 43] As the SQL generation prompt already contains 1,188 tokens, GPT might not be able to capture the entire concepts and the logic properly, resulting in the omission of the concepts in the query. Another

reason might be the performance changes of the OpenAI LLMs over time. Chen et al. assessed the performance of GPT models from March to June 2023 and observed significant performance variation. [17] Considering that our study period overlaps with their study period (from March to June 2023), the performance variation over time might influence our study result. Despite the current limitations, generative LLM's text-to-SQL performance is still impressive and can be beneficial. [28] The most frequent logic error was the missingness of the DISTINCT statement, which can be considered relatively not critical for cohort retrieval and can be revised with clear instructions in the criteria. Incorrect concept mapping is critical, but it was caused by the Usagi, which is not supported by LLM. However, the other logic-related errors such as missing values of the lab test or not specifying AND/OR in the SQL query might be critical, therefore require further investigation in future studies. Adopting a self-correction approach may reduce errors related to the validity of SQL queries. [44, 45] Although we did not systematically assess GPT's ability to self-correct SQL queries, LLM's self-correction has been shown to improve query accuracy in previous studies. [29, 46] Besides, except for syntax-related errors, the overall structures of GPT-generated SQL queries were concise and could be executed without human correction. Considering C2Q interface integrates a human-in-the-loop process, GPT-based semi-automated query generation, correction, and execution can be beneficial for gradually enhancing accuracy. Furthermore, rule-based query formulation which was used in C2Q 2.0 can be inherited in the future version allowing complementary with GPT-generated query. For example, we can further use a rule-based formulated query to fine-tune GPT or set it as an example query in the prompt.

Through the reasoning evaluation, we confirmed that GPT can help users understand GPT-generated queries and standardized concepts. Considering that some C2Q users are not familiar

with SQL or OMOP CDM and concept normalization, this narrative explanation can assist them in understanding how the system processes the given eligibility criteria. [21-23] However, the readability and correctness of the explanations achieved lower scores. Particularly, concept omissions, presentation of concept IDs, or inaccurate concept names hindered comprehension of the explanations. Nevertheless, the purpose of reasoning prompting is to provide users narrative description of how the system identifies the complex relations between the parsed clinical concepts and structured SQL queries to extract the cohort to prevent biased cohort extraction. As GPT can generate human-like text, users can clearly understand how the system defines the relation and logic between concepts and extracts data. Future studies should delve deeper into evaluating the interaction between human users and the outputs generated by systems using LLM.

This study has several limitations. First, as we are using OpenAI's API, we were unable to address certain issues such as model performance inconsistency and response time. Moreover, we observed that GPT occasionally did not follow prompt instructions and generated results in a random format. In the future, we are planning to deploy a local LLM to prevent performance variation over time. [47] Second, C2Q 3.0 requires improvements in user experience. The inclusion of the reasoning interface aimed to improve the interpretability of the system, but offering a narrative description could potentially lead to information overload for users. [48] Third, the number of clinical trials for evaluation is limited. Subsequent research should expand its scope by assessing more clinical trials with greater diversity.

CONCLUSION

This study demonstrated that C2Q 3.0 powered by OpenAI GPT-4 can significantly enhance the accuracy for concept extraction, query formulation, and reasoning of the eligibility criteria. Our approach, consisting of three GPT modules, can identify patient cohorts meeting the eligibility criteria while keeping patient records in the local environment without transferring OpenAI. Leveraging LLM might be not perfect, particularly regarding SQL generation and reasoning. Nevertheless, it has already shown a promising future, surpassing what traditional methods can achieve. Further investigation is required to confirm the stable and consistent use of generative LLM in the system.

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COMPETING INTERESTS

None declared.

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AUTHOR CONTRIBUTIONS

JP: Conceptualization, Methodology, Software Implementation, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Visualization, and Project administration. YF: Conceptualization, Methodology, Validation, Investigation, Writing –

Review & Editing, and Project administration. CT: Writing – Review & Editing. GZ: Writing – Review & Editing. BI: Methodology, Validation, Investigation, and Writing – Review & Editing. FC and DF: Investigation, and Writing – Review & Editing. RS: Methodology, Investigation, and Writing – Review & Editing. EG and MS: Investigation, and Writing – Review & Editing. CW: Conceptualization, Methodology, Validation, Resources, Writing – Review & Editing, Supervision, Project administration, and Funding acquisition.

FIGURE LEGENDS

Figure 1. Overall workflow of Criteria2Query 3.0 powered by GPT-4. The orange colored-arrows represent the interaction between human or GPT. If the users modify GPT-generated results, GPT will recognize the changes and re-execute each corresponding module.

Figure 2. Example architectures of three prompts in Criteria2Query 3.0

Figure 3. Manually identified SQL errors from GPT-generated queries on five clinical trials

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